

WEEK:- 12

Topic: OCR Technology, OCR.Space API Study, Image Processing, and Text Extraction

Objectives:

To understand Optical Character Recognition (OCR) technology, explore OCR.Space API functionality, perform image processing, and extract textual information from images and scanned documents efficiently.

Learning Outcomes:

- Learned the fundamentals and working principles of OCR technology.
- Understood OCR.Space API integration and implementation techniques.
- Gained knowledge of image preprocessing and enhancement methods.
- Developed skills in extracting and processing text from images.
- Improved understanding of automation and document digitization processes.

WORK SUMMARY:

During Week 12, I worked on OCR technology, OCR.Space API study, image processing, and text extraction. The training focused on understanding how OCR systems convert images and scanned documents into machine-readable text. I attended technical sessions and participated in practical exercises to explore the concepts, architecture, and real-world applications of OCR technology. Various document formats and image samples were analyzed to understand the challenges involved in text recognition and data extraction.

Throughout the week, I gained practical experience in integrating OCR.Space API and processing different types of image-based documents. I learned how image quality, resolution, and preprocessing techniques influence OCR accuracy and performance. Special emphasis was placed on image enhancement methods such as resizing, filtering, and noise reduction to improve text extraction results. I also explored techniques for handling extracted data and converting unstructured information into usable digital content.

The activities carried out during this week enhanced my understanding of intelligent document processing and automation technologies. Regular testing, validation exercises, and mentor feedback helped me improve my analytical and problem-solving skills. The knowledge and practical experience gained during this week provided a strong foundation for advanced OCR applications, document automation systems, and AI-based data extraction projects.

Activities Performed:

- Studied OCR technology and integrated OCR.Space API for text extraction.
- Performed image processing and analyzed OCR output accuracy.